



DEALER BEST PRACTICE SERIES

Improving The Protection Of Parts Using VCI Stretch Film

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

1.0	Introduction	1
2.0	Best Practice Description.....	1
3.0	Implementation Steps.....	3
4.0	Benefits	4
5.0	Resources Required	5
6.0	Supporting Attachments / References.....	5
7.0	Related Best Practices	5
8.0	Acknowledgements.....	5

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

A new method to improve the protection of parts from corrosion has been implemented at Machining Workshop in Ferreyros S.A. This product is called VCI (Vapor Corrosion Inhibitors) Stretch Film.

2.0 Best Practice Description

Before the use of VCI Stretch Film, the method of protecting parts after machining was to apply a liquid rust inhibitor then cover with shrink-wrap. After test and analysis of the cost versus benefit, the machining center now uses the VCI Stretch film. See Figure 1.



Figure 1. Products used before this best practice. 4A Rust Preventive (First) and Stretch Film without anticorrosive protection (Second)

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Improving The Protection Of Parts Using VCI Stretch
Film

DATE
2/19/2015

CHG
NO
00

NUMBER
1212-4.03-1299

Testing of the VCI Stretch Film as shown below in Figures 2 and 3 demonstrate that the material is very effective in the prevention of corrosion. By using the new film protection, it is not necessary to also use the 4A Rust Preventive because the VCI is incorporated into the VCI Stretch Film. Finally, this product is suitable for rust protection on iron based metals and products composed of different metals. The overall thickness of the film is 1 millimeter standard.

In the figure 2, half of the crankshaft was protected just with VCI Stretch Film.

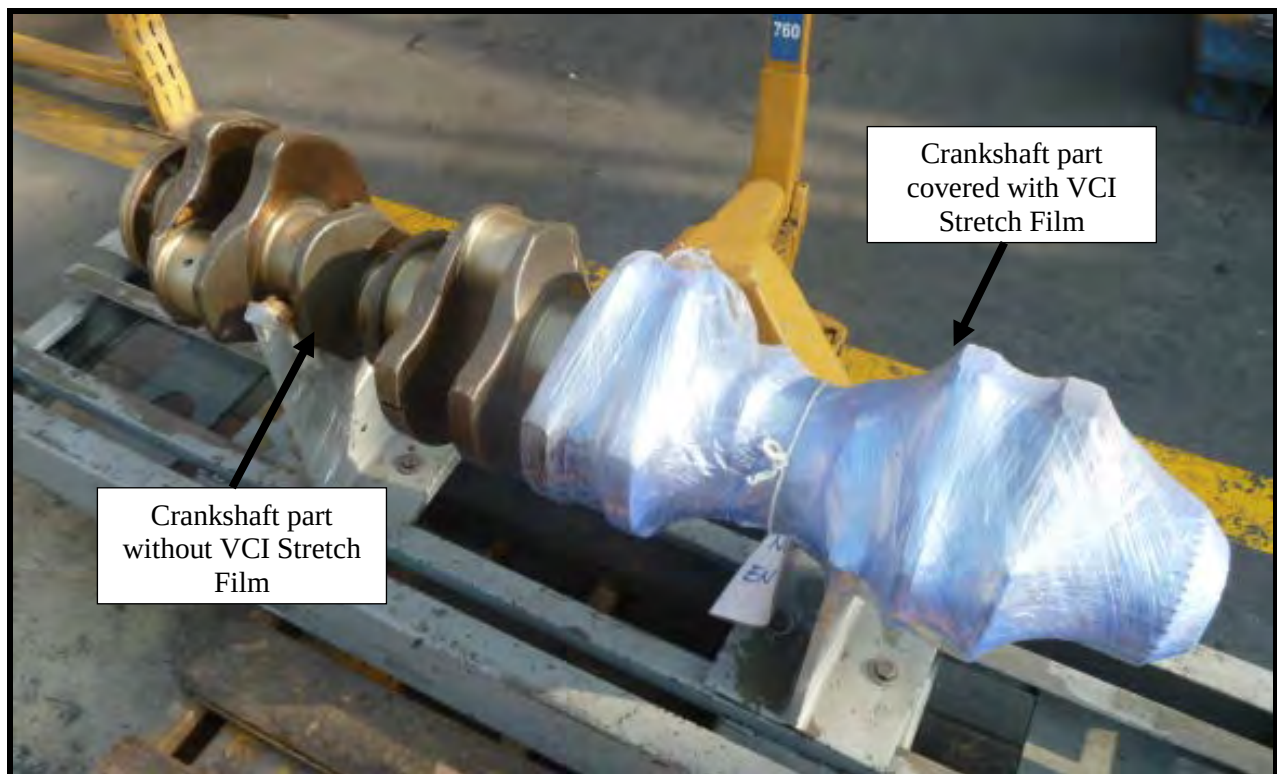


Figure 2. Example of crankshaft protected with VCI Stretch Film

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Improving The Protection Of Parts Using VCI Stretch
Film

DATE
2/19/2015

CHG
NO
00

NUMBER
1212-4.03-1299

After a time, the VCI Stretch Film was removed. The final results are showed in the Figure 3.

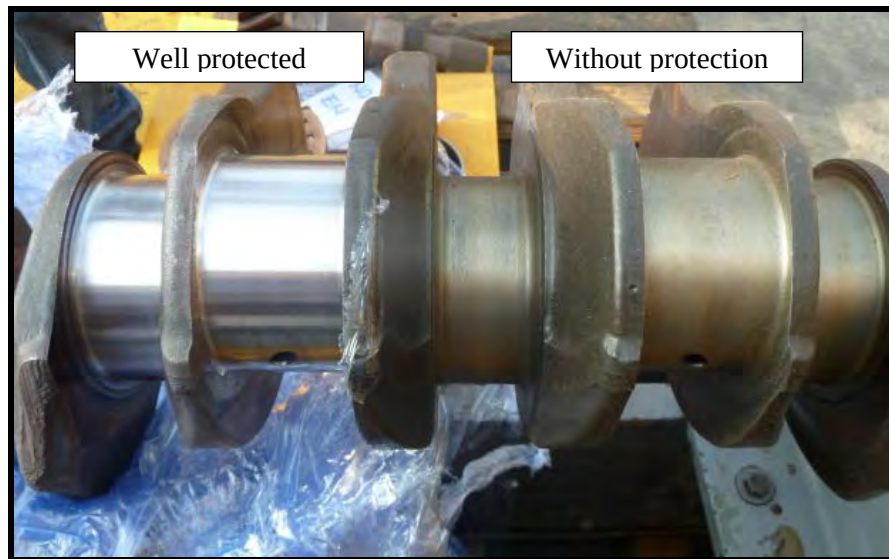


Figure 3. Crankshaft after removed of the VCI Stretch Film

Our testing showed very effective rust protection from the VCI Stretch Film.

3.0 Implementation Steps

- Acquire VCI Stretch Film – several brands exist and are easily found on the internet.



Figure 4. A crankshaft with VCI Stretch Film

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Improving The Protection Of Parts Using VCI Stretch
Film

DATE
2/19/2015

CHG
NO
00

NUMBER
1212-4.03-1299

4.0 Benefits

- Dual action: Contact and VCI protection
- Excellent clarity, cling, stretch and strength.
- No messy residues to remove
- Eliminate the use of 4A Rust Preventive and the Stretch Film without anticorrosive protection.
- Easy inspection and identification of shipping labels/bar codes (Work order).
- Reduce the cost of protection in 10.68% by year (See the table 1).

Item	Description	Unit	Cost h-h	Average parts/month	Average Total Cost	Total Cost Year 1	Total Cost Year 2	Total Cost Year 3	Total Cost
Materials									
1	Stretch Film sin VCI	m2	\$0.0572	22722.47	\$1,298.99	\$15,587.85	\$15,587.85	\$15,587.85	\$46,763.55
2	4A Rust Preventive	gallon	\$27.07	16.17	\$437.59	\$5,251.08	\$5,251.08	\$5,251.08	\$15,753.24
3	Elements of protection					\$1,002.48	\$1,002.48	\$1,002.48	\$3,007.44
4	Other consumables					\$300.00	\$300.00	\$300.00	\$900.00
Total Cost Materials						\$22,141.41	\$22,141.41	\$22,141.41	\$66,424.23
Item	Description	Unit	Cost h-h	Average parts/month	Average Total Cost	Total Cost Year 1	Total Cost Year 2	Total Cost Year 3	Total Cost
Labour force / Machine by reprocess									
1	Washing	Complete Works	\$12.50	300	\$3,750.00	\$45,000.00	\$45,000.00	\$45,000.00	\$135,000.00
2	SandBlasting	Complete Works	\$32.00	60	\$1920.00	\$23,040.00	\$23,040.00	\$23,040.00	\$69,120.00
3	Rectified / Machining	Complete Works	\$200.00	0.08	\$16.00	\$192.00	\$192.00	\$192.00	\$192.00
4	Inspector	Complete Works	\$7.00	100	\$700.00	\$8,400.00	\$8,400.00	\$8,400.00	\$25,200.00
Total Cost Labour force / Machine						\$ 76,632.00	\$ 76,632.00	\$ 76,632.00	\$ 229,896.00
Total Cost						\$ 98,773.41	\$ 98,773.41	\$ 98,773.41	\$ 296,320.23
Item	Description	Unit	Cost h-h	Average parts/month	Average Total Cost	Total Cost Year 1	Total Cost Year 2	Total Cost Year 3	Total Cost
Materials									
1	PMG VCI Stretch Film	m2	\$0.40	18177.98	\$7,351.57	\$88,218.82	\$88,218.82	\$88,218.82	\$264,656.46
Total Cost Materials						\$ 88,218.82	\$ 88,218.82	\$ 88,218.82	\$ 264,656.46
Direct Cost Savings						\$ 10,554.59	\$ 10,554.59	\$ 10,554.59	\$ 31,663.77

Table 1. Cost between Stretch Film without VCI vs VCI Stretch Film

The first four rows in the table above represent material costs of previous process using inhibitor and standard wrap.

The center rows are the costs to re-wash and clean parts when they are ready to be used.

The bottom of the table is the costs and savings using VCI Stretch Film.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Improving The Protection Of Parts Using VCI Stretch Film	DATE 2/19/2015	CHG NO 00	NUMBER 1212-4.03-1299

5.0 Resources Required

- Total Investment US\$ 7,351.57 for average monthly consumption of VCI Stretch Film's 26 rolls.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

This Best Practice was written by:

Name Frank Córdova R.

E mail frank.cordova@ferreyros.com.pe

Recognition:

Team: Miguel Mauriola / Operations Support Chief
Rafael Volta / Machine Chief
Liliana Falcón/Planning Chief
Jose Huayamis / Technique Communicator
Sponsor: Rodolfo Gerstein / Manager Machine Shop

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Improving The Protection Of Parts Using VCI Stretch Film	DATE 2/19/2015	CHG NO 00	NUMBER 1212-4.03-1299
---	-------------------	-----------------	--------------------------